

Quan Hua

Senior AI Engineer

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Professional Summary

Senior AI Engineer with 10+ years building production AI systems — from 3D body scanning at BodiData to autonomous robotics at Autonomous Inc. Most recently founded AIPriceAction, a full-stack AI financial analysis platform with a Rust backend, multi-agent research pipeline, and React SPA. Specialized in end-to-end ML from model design to deployment on edge devices and cloud.

Experience

Oct 2025 – Present: AIPriceAction - Creator & Lead Engineer

Website: aipriceaction.com GitHub: github.com/quanhua92/aipriceaction

- Built a Rust backend (Axum + PostgreSQL + Redis) with 22 workers syncing 574 tickers across Vietnamese stocks, crypto, and global markets.
- Designed a multi-agent research pipeline in Python (Supervisor → Workers → Aggregator → Reviewer) with automated quality gates.
- Published 3 installable AI Agent Skills for Claude Code, Gemini CLI, Cursor, and OpenCode.
- Shipped a React 19 + TypeScript SPA (17 routes, 47K+ lines) with TradingView-style charts and backtesting playground.
- Built a Python SDK (aipriceaction) and CLI (aipa-cli) with AI analysis across 13 templates.

2017 – Oct 2025: BodiData, Inc - <https://www.bodidata.com/> Senior Machine Learning Engineer

3D body measurement and size-matching solutions to reduce clothing returns. Led the CV engine across 3 generations of the Kora® 3D Body Scanner.

- **Kora V1 (Surface + RealSense):** Designed the CV pipeline from scratch — integrated ORB-SLAM 3 for real-time camera positioning, implemented the C++ application in Qt. Collaborated with a PhD researcher on the radar algorithm.
- **Kora V2 (RealSense D415):** Ported to far-range camera. Reduced measurement error to ~1 inch by improving 3D model fitting against fused point cloud data.
- **Kora V3 (iPad + ESP32 → AWS Lambda):** Rewrote ESP32 firmware to multi-threaded ESP-IDF with BLE to iPad and Vayyar radar integration for depth fusion.
- **Kora V3 Cloud:** Refactored Qt C++ processing pipeline into a CLI binary and built a Python Lambda to orchestrate S3 downloads, processing, and uploads.
- Rewrote a Node.js SSE streaming server in Rust (Axum + Tokio), eliminating OOM crashes and delivering 10x throughput on smaller ECS instances.

2014 – 2017: Autonomous, Inc. - <https://autonomous.ai/> Artificial Intelligence Engineer

- Led a team of 6 AI engineers building the deep learning brain for Maya Personal Robot — CES 2015 TechCrunch Hardware Battlefield finalist, \$161K+ raised on Kickstarter.
- Integrated and deployed deep learning models for computer vision, facial recognition, emotion detection, and NLP as ROS components with real-time inference on edge hardware.
- Owned the ML pipeline from data collection to production inference via TensorFlow Serving and Flask across multiple product lines.
- Developed prototypes for Smart Recycle Bin and Smart Receptionist, accelerating concept-to-demo cycles.

02/2014 – 03/2014: Research Internship Japan Advanced Institute of Science and Technology, Ishikawa, Japan

- Developed and tested a facial pose estimation algorithm using computer vision techniques on the BU-3DFE dataset, achieving competitive accuracy under varying head orientations.

Education

2010 – 2014: Bachelor of Science in Computer Science Honor Programs - University of Science Ho Chi Minh City. GPA: 8.7 / 10

Skills

- **Languages:** Rust, Python, C++, TypeScript
- **ML/DL:** PyTorch, Computer Vision, 3D Reconstruction
- **Frameworks:** LangChain, LangGraph, Qt, ESP-IDF, Axum, Tokio, React
- **Infrastructure:** PostgreSQL, Redis, Docker, AWS (Lambda, S3, ECS), CI/CD
- **AI Agents:** Multi-Agent Pipelines, Tool-Augmented LLMs, OpenAI API

Side Projects

UKYS: Great Fitting Shirts with Phone Measuring (Jun 2015 – Jan 2016)

Designed the core body measurement algorithm using Computer Vision, geometry, and camera models to enable iPhone/iPad users to measure their body in minutes for custom tailored shirts. **Won First Prize in “The Ambassador’s Entrepreneurship Challenge”** — presented by US Ambassador to Vietnam.

Publications

Book: Machine Learning with TensorFlow 1.x (2017) Co-authored 5 of 13 chapters on Deep Learning with TensorFlow.

Book: OpenCV3 Blueprints (2015) Co-authored 2 chapters: “Recognizing Facial Expressions with Machine Learning” and “Panoramic Image Stitching Application using Android Studio and NDK.”

2014: Hong-Quan Hua, T. Hoang Ngan Le, and Bac Le, “An Effective Initialization for ASM-based Methods”, CISIM 2014 Conference.